

Day 25 – HR Screening & AI Conversation Pipeline

Complete Deliverables Documentation

1. Project Objective

The objective of Day 25 was to develop and validate an **HR Screening and AI Conversation Pipeline** capable of understanding candidate answers and converting them into structured candidate information.

The pipeline performs:

- Intent classification
 - Answer understanding
 - Skill extraction
 - Skill integration
 - Experience extraction
 - Experience integration
 - Availability extraction
 - Availability integration
 - Salary extraction
 - Vague answer detection
 - Final integration
 - Final testing
-

2. Completed 12-Step Pipeline

Step Module		Status
1	Intent Classification	✔ Completed
2	Answer Understanding	✔ Completed
3	Skill Extraction	✔ Completed
4	Skill Integration	✔ Completed

Step	Module	Status
5	Experience Extraction	✓ Completed
6	Experience Integration	✓ Completed
7	Availability Extraction	✓ Completed
8	Availability Integration	✓ Completed
9	Salary Extraction	✓ Completed
10	Vague Answer Detection	✓ Completed
11	Final Integration	✓ Completed
12	Final Testing	✓ Completed

3. Availability Extraction Engine

Purpose

The Availability Extraction Engine identifies when a candidate can join an organization based on their HR screening answer.

Supported Availability Categories

- Immediate
- 15 days
- 30 days
- 2 weeks
- 1 week
- Notice period

Example

Candidate Answer:

"I can join within 15 days."

Extracted Availability:

15_days

Implementation

"""

Day 25 - Availability Extraction Engine

Purpose:

Extract candidate availability information
from HR screening answers.

"""

```
def extract_availability(answer: str):
```

```
    """
```

```
    Extract availability information from a candidate's answer.
```

```
    """
```

```
    if not answer or not answer.strip():
```

```
        return None
```

```
    text = answer.lower().strip()
```

```
    # Immediate availability
```

```
    immediate_keywords = [
```

```
        "immediately",
```

```
        "immediate",
```

```
        "right away",
```

```
        "available now",
```

```
        "join now",
```

```
]
```

```
if any(keyword in text for keyword in immediate_keywords):
```

```
    return "immediate"
```

```
# 15 days
```

```
if "15 days" in text or "15 day" in text:
```

```
    return "15_days"
```

```
# 30 days
```

```
if "30 days" in text or "30 day" in text:
```

```
    return "30_days"
```

```
# Two weeks
```

```
if "2 weeks" in text or "two weeks" in text:
```

```
    return "2_weeks"
```

```
# One week
```

```
if "1 week" in text or "one week" in text:
```

```
    return "1_week"
```

```
# Notice period
```

```
if "notice period" in text:
```

```
    return "notice_period"
```

```
return None
```

4. Availability Integration

Purpose

The Availability Integration module stores the extracted availability value inside the candidate's structured screening data.

Implementation

"""

Day 25 - Availability Integration

Purpose:

Integrate extracted candidate availability
into candidate screening data.

"""

```
def integrate_availability(candidate_data: dict, availability):
```

```
    """
```

```
    Integrate extracted availability into candidate data.
```

```
    """
```

```
    if candidate_data is None:
```

```
        candidate_data = {}
```

```
    if availability is not None:
```

```
        candidate_data["availability"] = availability
```

```
    return candidate_data
```

Example

Input:

```
candidate_data = {
```

```
"skills": ["Python", "FastAPI"],  
"experience_years": 2  
}
```

availability = "15_days"

Output:

```
{  
  "skills": ["Python", "FastAPI"],  
  "experience_years": 2,  
  "availability": "15_days"  
}
```

5. Salary Extraction Engine

Purpose

The Salary Extraction Engine extracts the expected salary from candidate answers.

It supports formats such as:

- 6 LPA
- 6.5 LPA
- 7 lakhs
- 8 lakh
- 10 lac

Implementation

"""

Day 25 - Salary Extraction Engine

Purpose:

Extract candidate salary expectation

from HR screening answers.

```
"""
```

```
import re
```

```
def extract_salary(answer: str):
```

```
    """
```

```
    Extract expected salary from a candidate's answer.
```

```
    """
```

```
    if not answer or not answer.strip():
```

```
        return None
```

```
    text = answer.lower().strip()
```

```
    match = re.search(
```

```
        r"(\d+(?:\.\d+)?)\s*(?:lpa|lakhs?|lac)",
```

```
        text
```

```
    )
```

```
    if match:
```

```
        return float(match.group(1))
```

```
    return None
```

Examples

"My expected salary is 6 LPA."

→ 6.0

"I expect 7.5 LPA."

→ 7.5

"My expectation is 8 lakhs."

→ 8.0

6. Vague Answer Detection

Purpose

The Vague Answer Detection module identifies unclear or non-specific candidate responses.

Examples of vague phrases

- Maybe
- Not sure
- I don't know
- Anything is fine
- Depends
- It depends
- Probably
- Perhaps
- I think
- Some experience

Implementation

""""

Day 25 - Vague Answer Detection

Purpose:

Detect vague or unclear answers

during HR screening.

""""


```
def detect_vague_answer(answer: str):  
    """  
    Detect whether a candidate's answer is vague.  
    """  
  
    if not answer or not answer.strip():  
        return False  
  
    text = answer.lower().strip()  
  
    vague_phrases = [  
        "maybe",  
        "not sure",  
        "i don't know",  
        "don't know",  
        "anything is fine",  
        "anything",  
        "depends",  
        "it depends",  
        "somewhere",  
        "a lot",  
        "good experience",  
        "some experience",  
        "i think",  
        "probably",  
        "perhaps",
```

```
]
```

```
if any(phrase in text for phrase in vague_phrases):
```

```
    return True
```

```
    return False
```

Examples

"Maybe I can join."

→ True

"It depends on the company."

→ True

"I have 2 years of experience."

→ False

"I can join within 15 days."

→ False

7. Final Integration

Purpose

The Final Integration module combines all extracted information into a single structured candidate screening result.

Implementation

""""

Day 25 - Final Screening Integration

Purpose:

Combine all extracted candidate information
into one structured screening result.

"""

```
def integrate_screening_result(
```

```
    answer,
```

```
    intent=None,
```

```
    skills=None,
```

```
    experience_years=None,
```

```
    availability=None,
```

```
    salary_lpa=None,
```

```
    is_missing=False,
```

```
    is_vague=False,
```

```
    is_off_topic=False
```

```
):
```

```
    """
```

Combine all screening information into one result.

```
    """
```

```
    return {
```

```
        "answer": answer,
```

```
        "intent": intent,
```

```
        "skills": skills if skills else [],
```

```
        "experience_years": experience_years,
```

```
        "availability": availability,
```

```
        "salary_lpa": salary_lpa,
```

```
        "is_missing": is_missing,
```

```
"is_vague": is_vague,  
"is_off_topic": is_off_topic  
}
```

Example Final Result

```
{  
  "answer": "I have 2 years of experience in Python.",  
  "intent": "experience",  
  "skills": ["Python"],  
  "experience_years": 2.0,  
  "availability": None,  
  "salary_lpa": None,  
  "is_missing": False,  
  "is_vague": False,  
  "is_off_topic": False  
}
```

8. Answer Understanding Validation

The Answer Understanding module was tested using multiple HR screening scenarios.

Test 1 – Experience

Answer:

I have 2 years of experience in Python.

Result:

```
{  
  'intent': 'experience',  
  'is_missing': False,  
  'is_vague': False,  
  'is_off_topic': False,  
  'skills': ['Python'],
```

```
'experience_years': 2.0,  
'availability': None  
}
```

Test 2 – Skills

Answer:

I am skilled in Python, FastAPI and Machine Learning.

Result:

```
{  
  'intent': 'skills',  
  'is_missing': False,  
  'is_vague': False,  
  'is_off_topic': False,  
  'skills': [  
    'Python',  
    'FastAPI',  
    'Machine Learning'  
  ],  
  'experience_years': None,  
  'availability': None  
}
```

Test 3 – Availability

Answer:

I can join immediately.

Result:

```
{  
  'intent': 'availability',  
  'is_missing': False,  
  'is_vague': False,
```

```
'is_off_topic': False,  
'skills': [],  
'experience_years': None,  
'availability': 'immediate'  
}
```

Test 4 – Salary Intent

Answer:

My expected salary is 6 LPA.

Result:

```
{  
  'intent': 'salary_expectation',  
  'is_missing': False,  
  'is_vague': False,  
  'is_off_topic': False,  
  'skills': [],  
  'experience_years': None,  
  'availability': None  
}
```

Test 5 – Introduction

Answer:

My name is Shinas and I am an AI Engineer.

Result:

```
{  
  'intent': 'introduction',  
  'is_missing': False,  
  'is_vague': False,  
  'is_off_topic': False  
}
```

Test 6 – Missing Answer

Answer:

Result:

```
{  
  'intent': 'missing_answer',  
  'is_missing': True,  
  'is_vague': False,  
  'is_off_topic': False  
}
```

Test 7 – Off-topic Answer

Answer:

I like watching movies.

Result:

```
{  
  'intent': 'unknown',  
  'is_missing': False,  
  'is_vague': False,  
  'is_off_topic': True  
}
```

9. Final Integration Testing

The final test file validated the complete integration layer.

Test Cases

Test Case	Result
Experience Answer	✅ PASSED
Skills Answer	✅ PASSED

Test Case	Result
Availability Answer	✅ PASSED
Salary Answer	✅ PASSED
Missing Answer	✅ PASSED
Off-topic Answer	✅ PASSED
Vague Answer	✅ PASSED

Verified Terminal Output

```
=====
FINAL INTEGRATION TEST
=====
Experience test: PASSED
Skills test: PASSED
Availability test: PASSED
Salary test: PASSED
Missing answer test: PASSED
Off-topic test: PASSED
Vague answer test: PASSED
=====
ALL FINAL INTEGRATION TESTS PASSED
=====
```

10. Final Architecture

The complete Day 25 pipeline follows this architecture:

Candidate Answer



Intent Classification



Answer Understanding



Skill Extraction



Skill Integration



Experience Extraction



Experience Integration



Availability Extraction



Availability Integration



Salary Extraction



Vague Answer Detection



Final Integration



Final Testing



Structured Candidate Screening Result

11. Final Structured Candidate Data

The completed system can produce a structured candidate screening object containing:

```
{  
  "answer": "...",
```

```
"intent": "...",  
"skills": [],  
"experience_years": None,  
"availability": None,  
"salary_lpa": None,  
"is_missing": False,  
"is_vague": False,  
"is_off_topic": False  
}
```

This structure can later be passed to the **candidate eligibility, scoring, ranking, and shortlisting layers**.

12. Day 25 Deliverables Summary

Deliverable	Status
Intent Classification	✓ Completed
Answer Understanding	✓ Completed
Skill Extraction	✓ Completed
Skill Integration	✓ Completed
Experience Extraction	✓ Completed
Experience Integration	✓ Completed
Availability Extraction	✓ Completed
Availability Integration	✓ Completed
Salary Extraction	✓ Completed
Vague Answer Detection	✓ Completed
Final Integration	✓ Completed

Deliverable	Status
Final Testing	 All Tests Passed

13. Conclusion

Day 25 – HR Screening & AI Conversation Pipeline has been successfully completed.

All **12 planned implementation steps** have been completed and validated.

The system can now:

- Understand candidate answers
- Identify answer intent
- Extract candidate skills
- Extract experience
- Extract availability
- Extract salary expectations
- Detect vague answers
- Detect missing answers
- Detect off-topic answers
- Integrate all extracted information
- Produce a structured candidate screening result
- Validate the complete integration through automated tests